Fiscal Year:	FY 2014	**Task Last Updated:**	FY 04/02/2014
PI Name:	Dantzig, Jonathan A Ph.D.		
Project Title:	Modeling Peritectic Microstructure Formation during Directional Solidification in Space and on Earth		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	dantzig@illinois.edu	**Fax:**	FY 217-244-6534
PI Organization Type:	UNIVERSITY	**Phone:**	217-333-4107
Organization Name:	University of Illinois, Urbana-Champaign		
PI Address 1:	352 Mechanical Engineering Building		
PI Address 2:	1206 West Green Street		
PI Web Page:	http://quattro.me.uiuc.edu/		
City:	Urbana	**State:**	IL
Zip Code:	61801	**Congressional District:**	15
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2009-10 Materials Science Rapid Turn Around Flight Opportunity
Start Date:	06/03/2010	**End Date:**	06/02/2015
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Downey, James1 Patton	**Contact Phone:**	
Contact Email:	patton.downey@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 6/02/2015 per NSSC information and PI (Ed., 4/3/14) NOTE: Period of performance changed to 6/3/2010-6/2/2014 (previously 4/6/10-4/5/14) per NSSC (Ed., 7/14/10)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX10AJ73G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Peritectic alloys constitute an important class of industrial alloys. The properties of these alloys depend strongly on their solidification history, and the potential exists to create novel materials with unique properties by controlling the solidification process. Peritectic alloys undergo an invariant reaction, in which a liquid phase l reacts with an existing solid phase alpha to form a second solid phase beta. When solidification occurs at very low growth rate in a high temperature gradient, this reaction can produce very complicated, non-equilibrium microstructures due to the limited diffusion in the solid state, and convection in the liquid. The European Space Agency (ESA) has an ongoing multi-national, multi-investigator program, called METCOMP, to study peritectic alloys. The proposed research will provide modeling support for one of these METCOMP projects, focusing on directional solidification in Cu-Sn alloys. Experiments performed on Earth are difficult to interpret because of the possibility of buoyancy-driven convection. Such experiments are necessary, however, to provide the scientific base for interpreting the space experiments. The objective of the proposed research is to develop and use models that do two things: (1) interact with the experimental work so as to improve the experiments themselves; and (2) provide detailed and predictive models to assist in interpretation of the experimental results. In order to accomplish the first objective, separate models for the macroscale transport phenomena and microstructure formation will be developed first. These models will then be merged into a single code that computes the transport and microstructure at the same time. This is feasible because the code will build on an existing code that uses adaptive mesh refinement to allocate computational resources efficiently.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research, if successful, will lead to improved products for peritectic alloys in general, and for Cu-Sn alloys in particular. There is experimental evidence that coupled growth in these systems can lead to novel microstructures with unique properties. However, these structures are difficult to produce and control This research could help resolve this problem, making these materials commercially available.
Task Progress:	The objective of this research is to provide modeling support for an ESA-funded experimental research program studying the development of coupled microstructures in directionally solidified (DS) Cu-Sn peritectic alloys. The project has two related tasks: (i) Use macroscale modeling to aid in the design of ground-based experiments that are free of convection due to solutal buoyancy; and (ii) Develop predictive microstructure models to assist in interpretation of the experimental results, defining the conditions that lead to coupled growth. Research in the first year on Task 1 showed that it is possible to achieve convection-free directional solidification (DS) of Cu-[14-21]wt% Sn alloys in a terrestrial environment by making the sample sufficiently small in diameter. This was first predicted in calculations performed under this grant, and then evidenced in experiments conducted by our ESA partners at EPFL. The ensuing work has focused on computational models for microstructure prediction in a convection-free environment. Experimental work at EPFL on DS of Cu-21wt% Sn alloys demonstrated that coupled lamellar growth of alpha-beta phases could be established over an extended range in the sample. The experiments were performed in s sufficiently small diameter tube that convection was effectively suppressed. These experiments evidenced several anomalous features: (i) The volume fraction of alpha phase, g, was substantially larger than expected from the equilibrium phase diagram; (ii) g oscillated with distance in the pulling direction z, with a period roughly equal to the lamellar spacing; and (iii) The mean volume fraction decreased slowly with distance from the onset of coupled growth. A series of computational models of varying complexity were developed to explain and understand these phenomena. This work has been presented at several conferences, and published in Acta Materialia. As part of this project we have also extended a previously developed adaptive phase-field code for binary alloy solidification to study the role of surface energy anisotropy in dendrite orientation selection. This work has also been presented at numerous conferences, and published in Metallurgical and Materials Transactions. Plans for Year 5 It has been speculated that the transition to coupled growth in peritectic alloys begins when the beta-phase nucleates on the alpha-phase at the interface, and then spreads rapidly across the surface. Further, the peritectic phase is unstable, breaking up into "fingers," which then become the lamellae observed during later growth. Thus, the length scale of the microstructure is set during this nucleation event. In this final year of the project, we will construct models of the early part of cooperative growth in order to establish the mechanism for setting the length scale of the microstructure. This calculation will require the extension of our previous studies into 3-D. The code has been installed, and the study of parameter space has been initiated.
Bibliography Type:	Description: (Last Updated: 04/03/2014)
Articles in Peer-reviewed Journals	Valloton J, Dantzig JA, Plapp M, Rappaz M. "Modeling of peritectic coupled growth in Cu-Sn alloys." Acta Materialia. 2013 Sep;61(15):5549-60. http://dx.doi.org/10.1016/j.actamat.2013.05.011 , Sep-2013
Articles in Peer-reviewed Journals	Dantzig JA, Di Napoli P, Friedli J, Rappaz M. "Dendritic growth morphologies in Al-Zn alloys - Part II: phase-field computations." Metallurgical and Materials Transactions A. 2013 Dec;44(12):5532-43. http://dx.doi.org/10.1007/s11661-013-1911-8 , Dec-2013